

MTG 207

Operations Management for Competitive Advantage

Instructor:	Danko Turcic, Associate Professor of Operations Management	Quarter:	Fall 2026
Office:	School of Business Building, 421	Meeting Time:	W 2:00 PM to 4:50 PM
Email:	dankot@ucr.edu	Classroom:	AGSM 118
		Office Hours (in person):	W 1:00 PM-2:00 PM (in person) Th 1:00 PM-2:00 PM (Zoom ID: 752 601 2800)

School of Business

Vision

We pursue excellence in all we do, as our vision is to be an internationally recognized leader in business education and research.

Mission

We create knowledge through impactful research, unlock potential through exceptional educational experiences, and foster success in an innovative and collaborative environment.

Strategy

We pursue excellence in all we do by cultivating our environment, engaging our community, and empowering our stakeholders.

Learning Goals

Problem Solving Skills

Students will be able to use a variety of theoretical perspectives to identify and critically evaluate the implications of business decisions for organizational stakeholders (e.g., customers, colleagues, employees, stockholders, suppliers, foreign governments, communities, cultures, and regulatory agencies) and the natural environment.

Professional Integrity / Ethical Reasoning Skills

Students will be able to recognize ethical issues, demonstrate familiarity with alternative frameworks for ethical reasoning, and discern trade-offs and implications of employing different ethical frames of reference when making business decisions.

Global Context Skills

Students will be conversant with major economic, social, political, and technological trends and conditions influencing foreign investment and development of the global economy and demonstrate an understanding of the cultural, interpersonal, and analytical competencies required for engaging in global business activities.

Written Communication Skills

Students will demonstrate proficiency in written communications by creating clearly written documents with appropriate content and conclusions.

Course Description

This paper focuses on managing the activities involved directly in creating products and services, such as design, production, and distribution. Provides managers with the skills and tools to analyze, optimize, and improve production processes for competitive advantage.

Learning Objectives

- Provide students with an understanding of operations management's role within organizations.
- Provide students with an understanding of how operations management can give firms a competitive advantage.
- Help students to become better problem solvers.

Course Materials

Required Text	Harvard Course Pack: See Canvas for instructions on how to purchase
Recommended Text	J. Heizer, B. Render, and C. Munson: Operations Management: Sustainability and Supply Chain Management, any recent edition
Lecture Notes	Electronic version of class notes will be periodically posted on Canvas
Additional References	B. Render, R.M. Stair Jr., and N. Balakrishnan, <i>Managerial Decision Modeling with Spreadsheets</i> , 2nd <u>or</u> 3rd edition, Prentice Hall

Grading Policy

Component	Weight
Cases analysis	50%
Final Case	20%
Homeworks	30%

Grades Scored Between		Letter Grade
97%	100%	A+
94%	97%	A
90%	94%	A-
87%	90%	B+
84%	87%	B
80%	84%	B-
77%	80%	C+
74%	77%	C
70%	74%	C-
67%	70%	D+
64%	67%	D
60%	64%	D-
0%	60%	F

Grades will be posted throughout the term on eLearn. If you find any problem with your score, you must inform the instructor within one week from the time this score is posted. After one week, scores will not be reviewed.

In the case of score dispute, the entire exam or assignment will be reviewed, not just the question(s) in dispute. The score could go up, down, or remain the same.

Case Studies

Each student must purchase a Harvard Business Publishing Course Pack. (The link to this course pack will be posted on Canvas.) We'll be running online simulations through the course pack, and only those who purchased it can receive credit for their work. The case studies will be done during regular class time, so class attendance will be essential.

Grading for Simulation-Based Cases

For case simulations where the primary objective is to maximize a performance metric (e.g., profit, throughput, market share), grades are assigned based on **relative class performance** using percentile ranks. This approach rewards deep understanding of the underlying operations concepts and encourages strategic thinking in a competitive setting.

Calculation of Grades

1. All submitted simulation results are collected and sorted in descending order by the target performance metric (e.g., final profit).
2. Each student's **percentile rank** is computed within the class (using inclusive percentile ranking, which fairly handles ties by assigning the same or interpolated rank to students with identical results).
3. Case scores (as percentages out of 100) are then assigned according to the following percentile thresholds:

Percentile Rank	Case Score (%)	Approximate Class Position
95–100	100	Top ~5%
90–94.9	95	Next ~5–10%
85–89.9	91	
80–84.9	88	
75–79.9	86	Middle to upper-middle
>0–74.9	80	
<i>No scores below 80% are assigned for submitted work.</i>		

Table 1: Scoring scale for simulation-based cases

Important notes:

- Students who submit a valid attempt (i.e., complete the required simulation runs and submit the requested deliverables) **will receive at least 80%**, regardless of their relative rank. This reflects that meaningful engagement with the material is rewarded.
- An **F (0%)** is assigned **only** in cases of non-submission, no attempt, or failure to follow submission instructions (e.g., missing files, wrong format, or no results provided).
- Ties are handled naturally: students with identical performance receive the same percentile rank and thus the same case score. No arbitrary tie-breakers are applied to the primary metric.
- When assigning final grades, the lowest three case scores will be dropped.

Final Case

One case will be assigned as a take-home case analysis. The grading procedure will be identical to the regular class cases (see “Calculation of Grades” above).

Homework

Homework provides an opportunity to practice the modeling and analysis skills introduced in the course. The homework will emphasize the quantitative aspects of the course material and provide feedback on how well you are mastering the analytical techniques. The best way to learn Operations Management is to work on problems. Homework will be posted in Canvas. Each posted homework will clearly show when it is due. Homework submissions will also be handled through Canvas. Each homework assignment allows three submission attempts per student, and only the final submitted attempt will be used for grading.

Academic Integrity

At the University of California, Riverside (UCR), honesty and integrity are fundamental values that guide and inform us as individuals and as a community. The academic culture requires students to take responsibility for learning and produce work reflecting their intellectual potential, curiosity, and capability. Students must represent themselves truthfully, claim only their own work, acknowledge their use of others' words, research results, and ideas, use the methods accepted by the appropriate academic disciplines, and engage honestly in all academic assignments. Misunderstanding of the appropriate academic conduct will not be accepted as an excuse for academic misconduct. If a student doubts appropriate academic conduct in a particular situation, he or she should consult with the instructor in the course to avoid a serious charge of academic misconduct. To ensure the highest standard of academic integrity, all students should be familiar with the guidelines posted at: <http://conduct.ucr.edu>

School of Business Guidelines Regarding Generative AI:

AI is not your work. Any work produced by AI that is passed off as your own is an act of academic dishonesty and violates UCR's academic honor code.

AI usage is subject to the discretion of each instructor. Each instructor will have specific guidelines on AI usage that must be adhered to for their specific class.

AI is not perfect. Its responses may contain inaccuracies, and it cannot critically evaluate the content it produces; you need to critically evaluate any output.

MGT 207 AI CLASS POLICY

The use of generative AI tools (e.g., ChatGPT, Gemini) is permitted for the following:

- Brainstorming and refining your ideas.
- Writing code
- Drafting an outline to organize your thoughts.

The use of generative AI tools is not permitted for the following:

- Generating entire answers to homework assignments (writing code to solve assignment problems using AI is okay)
- Generating answers on exams.

Course Schedule

WEDNESDAY	
Sep 30th	1
1 Using OM to Make Better Decisions	
1.1 Introduction	
• Introductions • Suggested Reading: §1 in J. Heizer, B. Render, and C. Munson: Operations Management: Sustainability and Supply Chain Management, any recent edition	
1.2 Case Study	
• “Advertising Budget”	
Oct 7th	2
2 Analytics Primer	
• Suggested Reading: Modules A and B included in Render, R.M. Stair Jr., and N. Balakrishnan, <i>Managerial Decision Modeling with Spreadsheets</i>, 3rd edition, Prentice Hall, §2.1-2.3, 2.7-2.8	
2.1 Case Study	
• “SimCase Simulation: Maketto” (see the HBP course pack)	
14th	3
3 Project Management	
• Suggested Reading: Modules B and F in J. Heizer, B. Render, and C. Munson: Operations Management: Sustainability and Supply Chain Management, any recent edition	
3.1 Case Study	
• Project Management Simulation: Scope, Resources, Schedule V3 (see the HBP course pack)	
21st	4
4 Forecasting	
• Suggested Reading: §4 in J. Heizer, B. Render, and C. Munson: Operations Management: Sustainability and Supply Chain Management, any recent edition	
4.1 Case Study	
• Data Analytics Simulation: Strategic Decision Making (see the HBP course pack)	

WEDNESDAY	
28th	5
5 Inventory Management • Suggested Reading: §12 and Module F in J. Heizer, B. Render, and C. Munson: Operations Management: Sustainability and Supply Chain Management, any recent edition 5.1 Case Study • “Operations Management Simulation: Inventory Basics” (see the HBP course pack)	
Nov 4th	6
6 Process and Capacity Management • Suggested Reading: §7 in J. Heizer, B. Render, and C. Munson: Operations Management: Sustainability and Supply Chain Management, any recent edition 6.1 Case Study • “Operations Management Simulation: Balancing Process Capacity” (see the HBP course pack)	
11th Veterans Day	
18th	7
7 Waiting Lines (Queues) • Suggested Reading: Modules D and F in J. Heizer, B. Render, and C. Munson: Operations Management: Sustainability and Supply Chain Management, any recent edition 7.1 Case Study • “Operations Management Simulation: System Utilization in Service Management” (see the HBP course pack)	
25th	8
8 Quality Control • Suggested Reading: §6 in J. Heizer, B. Render, and C. Munson: Operations Management: Sustainability and Supply Chain Management, any recent edition 8.1 Case Study • “Operations Management Simulation: Quality Analytics” (see the HBP course pack)	

9 Supply Chain Management

- **Suggested Reading:** §§11 and 12 in J. Heizer, B. Render, and C. Munson: Operations Management: Sustainability and Supply Chain Management, any recent edition

9.1 Case Study

- “Global Supply Chain Management Simulation V2” (see the HBP course pack)